



```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
```

<TableLayout

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:layout_editor_absoluteX="2dp"
    tools:layout_editor_absoluteY="93dp">

    <TableRow
        android:layout_gravity="center"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginLeft="5pt">

        <Button
            android:id="@+id/bt1"
            android:padding="0pt"

            android:textSize="50dp"
            android:layout_width="50pt"
            android:layout_height="50pt"
            android:layout_margin="3pt"
            android:onClick="btClick"
            app:backgroundTint="#BCB9B9" />

        <Button
            android:id="@+id/bt2"
            android:textSize="50dp"
            android:layout_width="50pt"
            android:layout_height="50pt"
            android:layout_margin="3pt"
            app:backgroundTint="#BCB9B9"
            android:onClick="btClick" />

        <Button
            android:id="@+id/bt3"
            android:layout_width="50pt"
            android:textSize="50dp"
            android:layout_height="50pt"
            app:backgroundTint="#BCB9B9"
            android:layout_margin="3pt"
            android:onClick="btClick" />
    </TableRow>

    <TableRow
        android:layout_gravity="center"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginLeft="5pt">
    >

    <Button
        android:id="@+id/bt4"
        android:textSize="50dp"
        android:layout_width="50pt"
        android:layout_height="50pt"
        app:backgroundTint="#BCB9B9"
        android:layout_margin="3pt"
```

```

        android:onClick="btClick" />

<Button
    android:id="@+id/bt5"
    android:textSize="50dp"
    android:layout_width="50pt"
    android:layout_height="50pt"
    android:layout_margin="3pt"
    app:backgroundTint="#BCB9B9"
    android:onClick="btClick" />

<Button
    android:id="@+id/bt6"
    android:layout_width="50pt"
    android:textSize="50dp"
    android:layout_height="50pt"
    android:layout_margin="3pt"
    app:backgroundTint="#BCB9B9"
    android:onClick="btClick" />
</TableRow>

<TableRow
    android:layout_gravity="center"
    android:layout_marginLeft="5pt"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Button
        android:id="@+id/bt7"
        android:layout_width="50pt"
        android:textSize="50dp"
        android:layout_height="50pt"
        android:layout_margin="3pt"
        app:backgroundTint="#BCB9B9"
        android:onClick="btClick" />

    <Button
        android:id="@+id/bt8"
        android:layout_width="50pt"
        android:textSize="50dp"
        android:layout_height="50pt"
        android:layout_margin="3pt"
        app:backgroundTint="#BCB9B9"
        android:onClick="btClick" />

    <Button
        android:id="@+id/bt9"
        android:layout_width="50pt"
        android:layout_height="50pt"
        android:layout_margin="3pt"
        android:textSize="50dp"
        app:backgroundTint="#BCB9B9"
        android:onClick="btClick" />
</TableRow>

<TableRow android:gravity="center"
    android:layout_margin="5pt">

```

```

        <Button
            android:id="@+id/btPlayAgain"
            android:padding="5pt"
            android:text="Bắt Đầu">

    </Button>
</TableRow>

<TableRow android:gravity="center"
    android:layout_margin="5pt">

    <TextView
        android:layout_span="3"
        android:id="@+id/txtShowresult"
        android:textSize="30sp"
        android:layout_width="match_parent"
        android:text="ÂÂ"
        android:gravity="center"
        android:visibility="invisible"
    ></TextView>
</TableRow>
</TableLayout>
</androidx.constraintlayout.widget.ConstraintLayout>

```

```
package com.hutech.tictoetoygame;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.graphics.Color;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.view.Window;
```

```
import android.widget.Button;
```

```
import android.widget.TextView;
```

```
import android.widget.Toast;
```

```
import java.util.ArrayList;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    int Winer = -1;
```

```
    int startGame=0;
```

```
    Button btPlayAgain, bt1, bt2, bt3, bt4, bt5, bt6, bt7, bt8, bt9;
```

```
    TextView txtShowresult;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        AnhXa();
```

```
        btPlayAgain.setOnClickListener(new View.OnClickListener() {
```

```
            @Override
```

```
            public void onClick(View v) {
```

```
                if (startGame == 1) {
```

```
                    PlayAgain();
```

```
                    startGame=0;
```

```

        btPlayAgain.setText("Bắt đầu");
    }
    else if (startGame == 0)
    {
        btPlayAgain.setText("Chơi lại");
        startGame=1;
    }
    }
    });
}

void PlayAgain() {
    Player1.clear();
    Player2.clear();
    Winer=-1;
    bt1.setText("");
    bt1.setBackgroundColor(Color.rgb(188,185,185));
    bt2.setText("");
    bt2.setBackgroundColor(Color.rgb(188,185,185));
    bt3.setText("");
    bt3.setBackgroundColor(Color.rgb(188,185,185));
    bt4.setText("");
    bt4.setBackgroundColor(Color.rgb(188,185,185));
    bt5.setText("");
    bt5.setBackgroundColor(Color.rgb(188,185,185));
    bt6.setText("");
    bt6.setBackgroundColor(Color.rgb(188,185,185));
    bt7.setText("");
    bt7.setBackgroundColor(Color.rgb(188,185,185));
    bt8.setText("");
    bt8.setBackgroundColor(Color.rgb(188,185,185));
    bt9.setText("");
    bt9.setBackgroundColor(Color.rgb(188,185,185));
    txtShowresult.setVisibility(View.INVISIBLE);
}

void AnhXa() {
    btPlayAgain = (Button) findViewById(R.id.btPlayAgain);
    bt1 = (Button) findViewById(R.id.bt1);
    bt2 = (Button) findViewById(R.id.bt2);
    bt3 = (Button) findViewById(R.id.bt3);
    bt4 = (Button) findViewById(R.id.bt4);
    bt5 = (Button) findViewById(R.id.bt5);
    bt6 = (Button) findViewById(R.id.bt6);
    bt7 = (Button) findViewById(R.id.bt7);
    bt8 = (Button) findViewById(R.id.bt8);
    bt9 = (Button) findViewById(R.id.bt9);
    txtShowresult = (TextView) findViewById(R.id.txtShowresult);
}

public void btClick(View view) {
    Button btSelected = (Button) view;

    int CellID=0;
    switch (btSelected.getId()) {
        case R.id.bt1:
            CellID = 1;

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        break;
    case R.id.bt2:
        CellID=2;
        break;
    case R.id.bt3:
        CellID=3;
        break;
    case R.id.bt4:
        CellID=4;
        break;
    case R.id.bt5:
        CellID=5;
        break;
    case R.id.bt6:
        CellID=6;
        break;
    case R.id.bt7:
        CellID=7;
        break;
    case R.id.bt8:
        CellID=8;
        break;
    case R.id.bt9:
        CellID=9;
        break;
    }

    if (Winer == -1 && startGame == 1)
    {
        PlayGame(CellID, btSelected);
    }

}

int ActivePlayer=1; // 1 : first, 2 second
ArrayList<Integer> Player1 = new ArrayList<Integer>();
ArrayList<Integer> Player2 = new ArrayList<Integer>();

void PlayGame(int CellID, Button btselected){

    if(ActivePlayer == 1)
    {
        btselected.setText("X");
        btselected.setBackgroundColor(Color.GREEN);
        btselected.setTextColor(Color.RED);
        Player1.add(CellID);
        ActivePlayer = 2;
    }
    else if (ActivePlayer ==2)
    {
        btselected.setText("O");
        btselected.setTextColor(Color.WHITE);
        btselected.setBackgroundColor(Color.BLUE);
        Player2.add(CellID);
        ActivePlayer = 1;
    }
}

```

```

    }

    CheckWiner();
    if (Winer == 1) {

        txtShowresult.setVisibility(View.VISIBLE);
        txtShowresult.setText("Player 1 thắng !");
        //Toast.makeText(this, "Play 1 thắng !", Toast.LENGTH_LONG).show();
    }
    else if (Winer == 2) {
        txtShowresult.setVisibility(View.VISIBLE);
        txtShowresult.setText("Player 2 thắng !");
        //Toast.makeText(this, "Play 2 thắng !", Toast.LENGTH_LONG).show();
    }
    else if (Winer == 0) {
        txtShowresult.setVisibility(View.VISIBLE);
        txtShowresult.setText("Hòa !");
        //Toast.makeText(this, "Hòa !", Toast.LENGTH_LONG).show();
    }
}

void CheckWiner(){

    //Dòng 1
    if(Player1.contains(1) && Player1.contains(2) &&
Player1.contains(3)){
        Winer =1;
    }
    if(Player2.contains(1) && Player2.contains(2) &&
Player2.contains(3)){
        Winer =2;
    }

    //Dòng 2
    if(Player1.contains(4) && Player1.contains(5) &&
Player1.contains(6)){
        Winer =1;
    }
    if(Player2.contains(4) && Player2.contains(5) &&
Player2.contains(6)){
        Winer =2;
    }

    //Dòng 3
    if(Player1.contains(7) && Player1.contains(8) &&
Player1.contains(9)){
        Winer =1;
    }
    if(Player2.contains(7) && Player2.contains(8) &&
Player2.contains(9)){
        Winer =2;
    }

    //Cột 1
    if(Player1.contains(1) && Player1.contains(4) &&
Player1.contains(7)){
        Winer =1;
    }
    if(Player2.contains(1) && Player2.contains(4) &&

```

```

Player2.contains(7)){
    Winer =2;
}

//Cột 2
if(Player1.contains(2) && Player1.contains(5) &&
Player1.contains(8)){
    Winer =1;
}
if(Player2.contains(2) && Player2.contains(5) &&
Player2.contains(8)){
    Winer =2;
}

//Cột 3
if(Player1.contains(2) && Player1.contains(5) &&
Player1.contains(8)){
    Winer =1;
}
if(Player2.contains(3) && Player2.contains(6) &&
Player2.contains(9)){
    Winer =2;
}

//Chéo 1
if(Player1.contains(1) && Player1.contains(5) &&
Player1.contains(9)){
    Winer =1;
}
if(Player2.contains(1) && Player2.contains(5) &&
Player2.contains(9)){
    Winer =2;
}

//Chéo 2
if(Player1.contains(3) && Player1.contains(5) &&
Player1.contains(7)){
    Winer =1;
}
if(Player2.contains(3) && Player2.contains(5) &&
Player2.contains(7)){
    Winer =2;
}

int sum = Player1.size() + Player2.size();
if (sum == 9 && Winer == -1)
{
    Winer = 0;
}

}
}

```